

CS312 Midterm Question Module No 01 to 11

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These questions should help you understand the core concepts and prepare effectively for your exam Happy studying!

- **### Module 1: Introduction to Databases**

1. What is the difference between data and information?
2. Describe the main components of a database system.
3. Explain the advantages of using a database management system (DBMS).

- **### Module 2: Database Environment**

1. What are the three levels of the database architecture?
2. Define and differentiate between logical and physical data independence.
3. Explain the roles of different types of users in a DBMS environment.

- **### Module 3: Data Modeling**

1. What is data modeling and why is it important in database design?
2. Explain the concept of an Entity-Relationship (ER) model.
3. Define and give examples of entities, relationships, and attributes.

- **### Module 4: Enhanced Entity-Relationship (EER) Model**

1. What is the difference between an ER model and an EER model?
2. Explain the concepts of generalization and specialization.
3. Describe the significance of aggregation in database modeling.

- **### Module 5: The Relational Model**

1. What are the fundamental concepts of the relational model?
2. Define and explain the significance of primary and foreign keys.
3. Describe the properties of relations in the relational model.

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<https://chat.whatsapp.com/Gec1rau2Nit0D4F5DjDiGM>

- **### Module 6: Relational Algebra**

1. What is relational algebra and why is it important?
2. Explain the different types of operations in relational algebra.
3. Give examples of selection and projection operations.

- **### Module 7: SQL Basics**

1. What is SQL and what are its main components?
2. Describe the difference between DDL, DML, and DCL in SQL.
3. Write an SQL query to create a table and insert data into it.

- **### Module 8: Advanced SQL**

1. Explain the concept of subqueries in SQL.
2. Describe the use of JOIN operations in SQL.
3. How can you enforce referential integrity using SQL constraints?

- **### Module 9: Database Design**

1. What are the steps involved in database design?
2. Explain the significance of normalization in database design.
3. Describe the different normal forms and their importance.

- **### Module 10: Fact Finding Techniques**

1. What are fact finding techniques and why are they important in database design?
2. Describe the use of interviews and questionnaires in gathering database requirements.
3. Explain the role of observation and document analysis in fact finding.

- **### Module 11: Three-Tier Architecture**

1. What is three-tier architecture and why is it important in database systems?
2. Describe the different layers in a three-tier architecture.
3. Explain the advantages of using three-tier architecture in a database environment.